# MAEG3080 Fundamentals of Machine Intelligence

CIFAR-100 · Custom PyTorch SE-ResNet · Final Project

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# Code & materials on GitHub

https://github.com/Rex-Yim/CIFAR100Classification

Scan the QR code or open the URL above: source code, Colab helpers, training scripts, the written report, and this slide deck (`main` branch).

# Goal

- Classify **CIFAR-100** with PyTorch *from scratch* (no pretrained weights)
- Compact **SE-ResNet** (squeeze-and-excitation attention)
- Pipeline: baseline $\rightarrow$ RandAugment finetune $\rightarrow$ Stage C (Mixup/CutMix/EMA)
- Final accuracy from course script `results.py`

# Dataset (quick facts)

| Scale |
| --- |
| 60k images · 32×32 RGB |
| 100 classes · 600 / class |

| Splits |
| --- |
| 50k train / 10k test |
| Finetune: 90% / 10% val (seed 42) |

# Model

- `se_resnet`: residual blocks + **SE** channel attention
- Base width 32 · 2 blocks per stage · GAP $\rightarrow$ 100-way linear

# Three-stage training

**Stage A** 4 epochs · basic aug · ∼33% test checkpoint

**Stage B** Init from Stage A · RandAugment · 30 epochs · val selection

**Stage C** From Stage B `best.pt` · Mixup/CutMix · EMA · lower LR

`scripts/iterate_from_stageb_stage_c_mixema.sh` · Colab · AMP

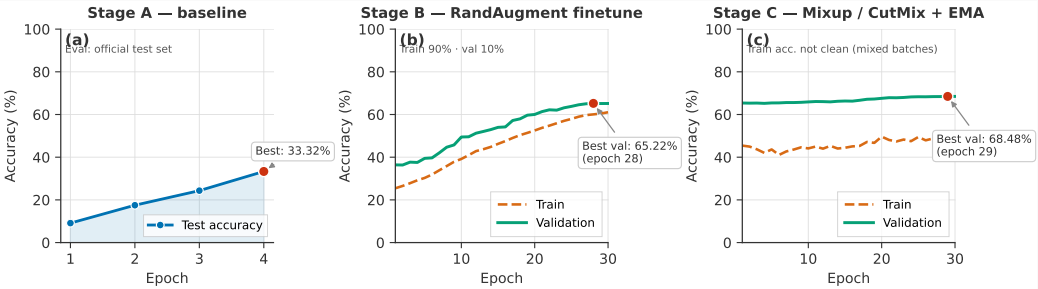
# Results

| Setting | Accuracy |
|---|---|
| Stage A baseline (4 ep., test) | $\approx 33.3\%$ |
| Stage B official test | **65.17%** |
| Stage C official test | **68.14%** |

**Best model**

Stage C checkpoint `sota_se_resnet_exp1/best.pt` · evaluated with `results.py`

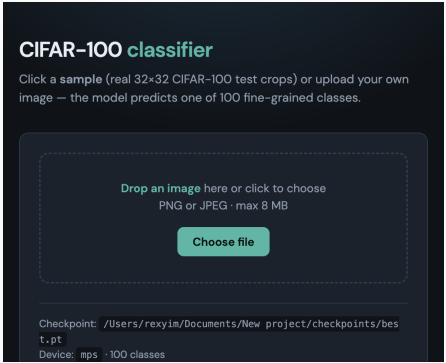
# Training curves

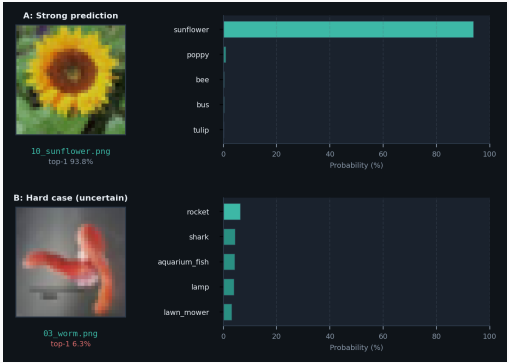



(a) Test · (b) train vs val · (c) Stage C (mixed-batch train acc. not directly comparable)

# Interactive demo (local)

- `demo/` FastAPI: upload → top-$k$ classes (SE-ResNet + `best.pt`)
- Run: `uvicorn demo.app:app --port 8765` · `http://127.0.0.1:8765`




*Right:* auto-picked from shipped samples — **strong** (highest top-1 conf.) vs. **weak** (lowest top-1); filenames under crops; same checkpoint as UI.

# What worked

- Stable Stage A init before heavy augmentation
- RandAugment + longer schedule (Stage B $\approx$ 65% test)
- Stage C on top of Stage B: Mixup/CutMix + EMA ($\approx$ 68% test)

# Challenges

- CIFAR-100 is fine-grained and small-input hard
- Long / huge models costly on free Colab
- SE-ResNet: faster iteration vs. lower ceiling than giant WRNs

# Contribution

- Rex Yim: data pipeline, model implementation, training runs, Colab workflow, report, and slide preparation
- Jason Chan: brainstorming ideas, preprocessing discussions, epoch-size planning, and support for presentation and Q&A practice
- Tools: Codex $\rightarrow$ Cursor · GPU on Colab

# Takeaway

**68.14%** official test (Stage C) | **65.17%** (Stage B only)

Questions?

Project repository: `https://github.com/Rex-Yim/CIFAR100Classification`